

MATRICES

1.	If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $(3I + 4A)(3I - 4A) = x^2I$, then the value(s) x is/are : (a) $\pm \sqrt{7}$ (b) 0 (c) ± 5 (d) 25
2.	If for a square matrix A, $A^2 - 3A + I = O$ and $A^{-1} = xA + yI$, then the value of $x + y$ is : (a) -2 (b) 2 (c) 3 (d) -3
3.	If $A = \begin{bmatrix} 3 & 4 \\ 5 & 2 \end{bmatrix}$ and $2A + B$ is a null matrix, then B is equal to : (a) $\begin{bmatrix} 6 & 8 \\ 10 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} -6 & -8 \\ -10 & -4 \end{bmatrix}$ (c) $\begin{bmatrix} 5 & 8 \\ 10 & 3 \end{bmatrix}$ (d) $\begin{bmatrix} -5 & -8 \\ -10 & -3 \end{bmatrix}$
4.	If A is a 2×3 matrix such that AB and AB' both are defined, then order of the matrix B is (A) 2×2 (B) 2×1 (C) 3×2 (D) 3×3
5.	If $\begin{bmatrix} 2 & 0 \\ 5 & 4 \end{bmatrix} = P + Q$, where P is a symmetric and Q is a skew symmetric matrix, then Q is equal to (A) $\begin{bmatrix} 2 & 5/2 \\ 5/2 & 4 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 5/2 \\ -5/2 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & -5/2 \\ 5/2 & 4 \end{bmatrix}$

6.	If $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 3 & a & 1 \end{bmatrix}$ is non-singular matrix and $a \in A$, then the set A is (A) $\mathbb{R}$ (B) $\{0\}$ (C) $\{4\}$ (D) $\mathbb{R} - \{4\}$
7.	If $f(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then prove that $f(\alpha) \cdot f(-\beta) = f(\alpha - \beta)$
8.	A and B are square matrices of same order. If $(A + B)^2 = A^2 + B^2$, then : (a) $AB = BA$ (b) $AB = -BA$ (c) $AB = O$ (d) $BA = O$
9.	A and B are skew-symmetric matrices of same order. AB is symmetric, if (a) $AB = O$ (b) $AB = -BA$ (c) $AB = BA$ (d) $BA = O$
10.	For what value of $x \in \left[0, \frac{\pi}{2}\right]$, is $A + A' = \sqrt{3} I$, where $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$? (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) 0 (d) $\frac{\pi}{2}$
11.	If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that $A^3 - 6A^2 + 7A + 2I = O$.
12.	If $x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$, then : (a) $x = 1, y = 2$ (b) $x = 2, y = 1$ (c) $x = 1, y = -1$ (d) $x = 3, y = 2$

13.

The product $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ is equal to :

(a) $\begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix}$

(b) $\begin{bmatrix} (a+b)^2 & 0 \\ (a+b)^2 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} a^2 + b^2 & 0 \\ a^2 + b^2 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$

14.

If A is a square matrix and $A^2 = A$, then $(I + A)^2 - 3A$ is equal to :

(a) I

(b) A

(c) 2A

(d) 3 I

15.

If a matrix $A = [1 \ 2 \ 3]$, then the matrix AA' (where A' is the transpose of A) is :

(a) 14

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$

(d) [14]

16.

If $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$ and $A = B^2$, then x equals

(a) ± 1

(b) -1

(c) 1

(d) 2

17.	If $A = [a_{ij}]$ is a square matrix of order 2 such that $a_{ij} = \begin{cases} 1, & \text{when } i \neq j \\ 0, & \text{when } i = j \end{cases}$, then A^2 is (a) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
18.	For two matrices A and B, given that $A^{-1} = \frac{1}{4}B$, then inverse of $(4A)$ is : (A) $4B$ (B) B (C) $\frac{1}{4}B$ (D) $\frac{1}{16}B$
19.	If X, Y and XY are matrices of order 2×3, $m \times n$ and 2×5 respectively, then number of elements in matrix Y is : (A) 6 (B) 10 (C) 15 (D) 35
20.	(a) Find a matrix A such that $A \begin{bmatrix} 4 & 0 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 10 \\ 0 & -16 \end{bmatrix}.$ Also, find A^{-1}.
21.	Given a square matrix A of order 3 such that $A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$, show that $A^3 = A^{-1}$.
22.	If A is a square matrix of order 3 and $A = 6$, then the value of $\text{adj } A$ is : (a) 6 (b) 36 (c) 27 (d) 216

23.	If matrix $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $A^2 = kA$, then the value of k is : (a) 1 (b) -2 (c) 2 (d) -1
24.	It is given that $X \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Then matrix X is : (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$
25.	If a matrix has 36 elements, the number of possible orders it can have, is : (A) 13 (B) 3 (C) 5 (D) 9
26.	If $\begin{bmatrix} x+y & 2 \\ 5 & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$, then the value of $\left(\frac{24}{x} + \frac{24}{y}\right)$ is : (A) 7 (B) 6 (C) 8 (D) 18
27.	If A and B are two non-zero square matrices of same order such that $(A + B)^2 = A^2 + B^2$, then : (A) $AB = O$ (B) $AB = -BA$ (C) $BA = O$ (D) $AB = BA$
28.	Assertion (A) : For matrix $A = \begin{bmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 1 \end{bmatrix}$, where $\theta \in [0, 2\pi]$, $A \in [2, 4]$. Reason (R) : $\cos \theta \in [-1, 1], \forall \theta \in [0, 2\pi]$.

29.	If $A = \begin{bmatrix} -1 & a & 2 \\ 1 & 2 & x \\ 3 & 1 & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ -8 & 7 & -5 \\ b & y & 3 \end{bmatrix}$, find the value of $(a + x) - (b + y)$.
30.	The number of all scalar matrices of order 3, with each entry $-1, 0$ or 1, is : (A) 1 (B) 3 (C) 2 (D) 3^9
31.	If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $[F(x)]^2 = F(kx)$, then the value of k is : (A) 1 (B) 2 (C) 0 (D) -2
32.	If the sum of all the elements of a 3×3 scalar matrix is 9, then the product of all its elements is : (A) 0 (B) 9 (C) 27 (D) 729
33.	<i>Assertion (A) :</i> For any symmetric matrix A, $B'AB$ is a skew-symmetric matrix. <i>Reason (R) :</i> A square matrix P is skew-symmetric if $P' = -P$.

34.	If $A = [a_{ij}]$ is an identity matrix, then which of the following is true ? (A) $a_{ij} = \begin{cases} 0, & \text{if } i = j \\ 1, & \text{if } i \neq j \end{cases}$ (B) $a_{ij} = 1, \forall i, j$ (C) $a_{ij} = 0, \forall i, j$ (D) $a_{ij} = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$
35.	Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a square matrix such that $\text{adj } A = A$. Then, $(a + b + c + d)$ is equal to : (A) $2a$ (B) $2b$ (C) $2c$ (D) 0
36.	If A and B are two skew symmetric matrices, then $(AB + BA)$ is : (A) a skew symmetric matrix (B) a symmetric matrix (C) a null matrix (D) an identity matrix
37.	<i>Assertion (A) :</i> Every scalar matrix is a diagonal matrix. <i>Reason (R) :</i> In a diagonal matrix, all the diagonal elements are 0.
38.	If $\begin{bmatrix} a & c & 0 \\ b & d & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is a scalar matrix, then the value of $a + 2b + 3c + 4d$ is : (A) 0 (B) 5 (C) 10 (D) 25

39.

Given that $A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$, matrix A is :

(A) $7 \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(C) $\frac{1}{7} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

(D) $\frac{1}{49} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

40.

If $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$, then the value of $I - A + A^2 - A^3 + \dots$ is :

(A) $\begin{bmatrix} -1 & -1 \\ 4 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

41.

Given that $[1 \quad x] \begin{bmatrix} 4 & 0 \\ -2 & 0 \end{bmatrix} = 0$, the value of x is :

(A) -4

(B) -2

(C) 2

(D) 4

42.

If $A = \begin{bmatrix} a & c & -1 \\ b & 0 & 5 \\ 1 & -5 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the value of $2a - (b + c)$ is :

(A) 0

(B) 1

(C) -10

(D) 10

43.	If A is a square matrix of order 3 such that the value of $\text{adj} \cdot A = 8$, then the value of A^T is : (A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 8 (D) $2\sqrt{2}$
44.	If inverse of matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 1 & \lambda & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then value of λ is : (A) -4 (B) 1 (C) 3 (D) 4
45.	If $[x \ 2 \ 0] \begin{bmatrix} 5 \\ -1 \\ x \end{bmatrix} = [3 \ 1] \begin{bmatrix} -2 \\ x \end{bmatrix}$, then value of x is : (A) -1 (B) 0 (C) 1 (D) 2
46.	Find the matrix A^2, where $A = [a_{ij}]$ is a 2×2 matrix whose elements are given by $a_{ij} = \text{maximum}(i, j) - \text{minimum}(i, j)$: (A) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
47.	Let $A = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 4 & -1 \\ -3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -2 \\ -5 \\ -7 \end{bmatrix}$, $C = [9 \ 8 \ 7]$, which of the following is defined ? (A) Only AB (B) Only AC (C) Only BA (D) All AB, AC and BA

48.	If $A = \begin{bmatrix} 7 & 0 & x \\ 0 & 7 & 0 \\ 0 & 0 & y \end{bmatrix}$ is a scalar matrix, then y^x is equal to (A) 0 (B) 1 (C) 7 (D) ± 7
49.	Let P be a skew-symmetric matrix of order 3. If $\det(P) = \alpha$, then $(2025)^\alpha$ is (A) 0 (B) 1 (C) 2025 (D) $(2025)^3$
50.	Four friends Abhay, Bina, Chhaya and Devesh were asked to simplify $4AB + 3(AB + BA) - 4BA$, where A and B are both matrices of order 2×2. It is known that $A \neq B \neq I$ and $A^{-1} \neq B$. Their answers are given as : Abhay : $6AB$ Bina : $7AB - BA$ Chhaya : $8AB$ Devesh : $7BA - AB$ Who answered it correctly ? (A) Abhay (B) Bina (C) Chhaya (D) Devesh
51.	Which of the following can be both a symmetric and skew-symmetric matrix ? (A) Unit Matrix (B) Diagonal Matrix (C) Null Matrix (D) Row Matrix
52.	If A and B are square matrices of order m such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following is always correct ? (A) $A = B$ (B) $AB = BA$ (C) $A = 0$ or $B = 0$ (D) $A = I$ or $B = I$
53.	Assertion (A) : $A = \text{diag} [3 \ 5 \ 2]$ is a scalar matrix of order 3×3. Reason (R) : If a diagonal matrix has all non-zero elements equal, it is known as a scalar matrix.

54.	The matrix $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & -7 \\ 2 & 7 & 0 \end{bmatrix}$ is a : (A) diagonal matrix (B) symmetric matrix (C) skew symmetric matrix (D) scalar matrix
55.	If A and B are square matrices of same order such that $AB = A$ and $BA = B$, then $A^2 + B^2$ is equal to : (A) $A + B$ (B) BA (C) $2(A + B)$ (D) $2BA$
56.	If $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ and $A^2 = kA$, then find the value of k.
57.	If $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$, then A^3 is : (A) $3 \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ (B) $\begin{bmatrix} 125 & 0 & 0 \\ 0 & 125 & 0 \\ 0 & 0 & 125 \end{bmatrix}$ (C) $\begin{bmatrix} 15 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 15 \end{bmatrix}$ (D) $\begin{bmatrix} 5^3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$
58.	If $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 3 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ -1 & 2 \\ 0 & 5 \end{bmatrix}$, then the correct statement is : (A) Only AB is defined. (B) Only BA is defined. (C) AB and BA, both are defined. (D) AB and BA, both are not defined.

59.	If $\begin{bmatrix} 4+x & x-1 \\ -2 & 3 \end{bmatrix}$ is a singular matrix, then the value of x is : (A) 0 (B) 1 (C) -2 (D) -4
60.	Sum of two skew-symmetric matrices of same order is always a/an : (A) skew-symmetric matrix (B) symmetric matrix (C) null matrix (D) identity matrix
61.	If $A = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 0 & 5 \\ -8 & -5 & 0 \end{bmatrix}$, then A is a : (A) null matrix (B) symmetric matrix (C) skew-symmetric matrix (D) diagonal matrix
62.	Let both AB' and $B'A$ be defined for matrices A and B. If order of A is $n \times m$, then the order of B is : (A) $n \times n$ (B) $n \times m$ (C) $m \times m$ (D) $m \times n$
63.	If $A = \begin{bmatrix} 1 & -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 4 \\ 0 & 5 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$, are three matrices, then find ABC.
64.	What is the total number of possible matrices of order 3×3 with each entry as $\sqrt{2}$ or $\sqrt{3}$? (A) 9 (B) 512 (C) 615 (D) 64

65. The matrix $A = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{5} \end{bmatrix}$ is a/an :

- (A) scalar matrix (B) identity matrix
(C) null matrix (D) symmetric matrix

66. If $\begin{bmatrix} 2x-1 & 3x \\ 0 & y^2-1 \end{bmatrix} = \begin{bmatrix} x+3 & 12 \\ 0 & 35 \end{bmatrix}$, then the value of $(x-y)$ is :

- (A) 2 or 10
- (B) - 2 or 10
- (C) 2 or - 10
- (D) - 2 or - 10

67. If $A = \begin{bmatrix} 1 & 0 \\ -1 & 5 \end{bmatrix}$, then find the value of K if $A^2 = 6A + KI_2$, where I_2 is an identity matrix.

CASE STUDY

A scholarship is a sum of money provided to a student to help him or her pay for education. Some students are granted scholarships based on their academic achievements, while others are rewarded based on their financial needs.



Every year a school offers scholarships to girl children and meritorious achievers based on certain criteria. In the session 2022 – 23, the school offered monthly scholarship of ₹ 3,000 each to some girl students and ₹ 4,000 each to meritorious achievers in academics as well as sports.

In all, 50 students were given the scholarships and monthly expenditure incurred by the school on scholarships was ₹ 1,80,000.

Based on the above information, answer the following questions :

- (i) Express the given information algebraically using matrices.
- (ii) Check whether the system of matrix equations so obtained is consistent or not.
- (iii) (a) Find the number of scholarships of each kind given by the school, using matrices.
- (b) Had the amount of scholarship given to each girl child and meritorious student been interchanged, what would be the monthly expenditure incurred by the school ?

69. Gautam buys 5 pens, 3 bags and 1 instrument box and pays a sum of ₹ 160. From the same shop, Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays a sum of ₹ 190. Also Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays a sum of ₹ 250.

Based on the above information, answer the following questions :

(I) Convert the given above situation into a matrix equation of the form $AX = B$.

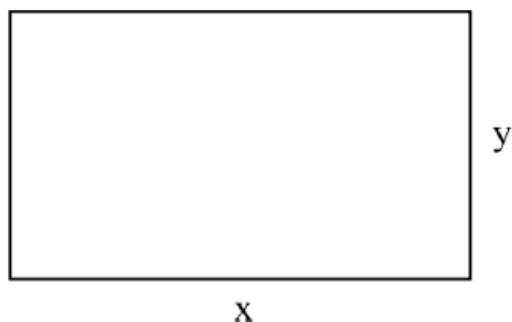
(II) Find $|A|$.

(III) Find A^{-1} .

OR

(III) Determine $P = A^2 - 5A$.

70. An architect is developing a plot of land for a commercial complex. When asked about the dimensions of the plot, he said that if the length is decreased by 25 m and the breadth is increased by 25 m, then its area increases by 625 m^2 . If the length is decreased by 20 m and the breadth is increased by 10 m, then its area decreases by 200 m^2 .



On the basis of the above information, answer the following questions :

- (i) Formulate the linear equations in x and y to represent the given information.
- (ii) Find the dimensions of the plot of land by matrix method.